PII Detection and Masking

Fine-Tuned RoBERTa vs. Zero-Shot LLaMA 1B

1. Approach

1.1 Dataset Preparation

The base WikiNeural dataset (28,516 train / 3,650 test entries) contained only person-name labels (B-PER, I-PER). Synthetic emails were injected into 15% of entries containing person names using two strategies:

- **Normal emails (~70%):** single-token addresses like mweiner@nfl.com tagged B-EMAIL.
- **Spaced emails (~30%):** multi-token addresses like anwar37 @ bbc . co . uk tagged B-EMAIL (first) + I-EMAIL (rest).

Train and test use completely disjoint sentence template sets (15 each) to prevent data leakage. Single-token names are handled to avoid unrealistic patterns like madonna.madonna@…

1.2 Tokenization & Split

RoBERTa tokenizer (roberta-base) with max_length=256. A critical fix assigns I-EMAIL to sub-tokens of email tokens (e.g., john.doe@gmail.com → john, ., doe, @, gmail, ., com each get B-EMAIL/I-EMAIL). Stratified 90/10 split preserves the 15% email ratio. 800 sequences (2.5%) exceeded 256 sub-tokens.

1.3 Fine-Tuning Configuration

roberta-base (124M params) fine-tuned for 5 epochs on T4 GPU: batch 16 × 2 accumulation = effective 32, lr = 2e-5, warmup = 401 steps, FP16 + gradient checkpointing, early stopping (patience 2). Best model selected by validation F1. Evaluation uses seqeval (span-level P/R/F1) and sklearn (token-level FPR/FNR).

1.4 Zero-Shot LLM Setup

meta-llama/Llama-3.2-1B-Instruct (float16) evaluated zero-shot with instruction-only prompt (no few-shot examples). Outputs JSON with extracted names and emails. Evaluated on a 325-example stratified subset (250 name-only, 75 with emails). Metrics computed at both sentence-level and entity-level.

2. Results

2.1 RoBERTa — Test Set Evaluation

Entity	Precision	Recall	F1 (span)	FPR	FNR
PER	0.9282	0.9332	0.9307	0.0033	0.0619
EMAIL	0.9380	0.9607	0.9492	0.0002	0.0393

Total errors: 584 / 98,497 tokens (0.59%). Cross-entity confusions: 0. EMAIL FP: 1, EMAIL FN: 0.

2.2 LLaMA 1B — Zero-Shot (325-example subset)

Entity	P (entity)	R (entity)	F1 (entity)	P (sentence)	R (sentence)
PER	0.9084	0.8038	0.8529	1.0000	0.8215
EMAIL	0.2771	0.8533	0.4183	0.3524	0.9867

Hallucinations: 136/325 (41.8%). Bad JSON: 1. Spaced email format mismatches contributed to EMAIL FP.

2.3 Comparison

Metric	RoBERTa (fine-tuned)	LLaMA 1B (zero-shot)	Delta
PER F1	0.9307	0.8529	+0.0778
EMAIL F1	0.9492	0.4183	+0.5309
Hallucinations	0	136/325 (41.8%)	—
Bad output format	0	1	—

RoBERTa outperforms LLaMA 1B by 8–53 percentage points across all metrics, with the gap most pronounced for EMAIL detection (53 pp).

3. Error Analysis

■ RoBERTa Errors (584 total / 98,497 tokens)

- **PER FP (291):** Named entities in non-person contexts — 'Le' (restaurant), 'Daniel' (biblical book), 'IIR' (filter type). Model over-generalises capitalisation + position cues.
- **PER FN (292):** Names in quoted/compound contexts — 'Joan of Arc', 'Nemesis' (ride), 'Mounir El Hamdaoui'. Multi-word or unusual names in non-standard positions.
- **EMAIL FP (1):** A single '.' in '42.77 %' tagged as I-EMAIL — artifact of spaced email training pattern.
- **EMAIL FN (0):** Perfect recall on all test emails.

■ LLaMA 1B Errors

- **Email hallucinations (136):** Fabricated emails like drake@starboyapp.com and rihanna@rihanna.com in sentences with no emails. Also output empty strings as emails.
- **PER name misses (58):** Less common names or names in complex sentences were missed (recall 80.4%).
- **Spaced format mismatch:** LLaMA returned 'matisyahu58 @ hotmail. com' (with spaces) which didn't match reconstructed ground truth, inflating EMAIL FP count.

4. Challenges

- **Hallucination is the dominant failure mode.** LLaMA 1B invented plausible-looking but fictitious emails in 41.8% of non-email examples. Without supervised training on the target schema, the model has no calibration for when NOT to predict.
- **Format inconsistency.** The model outputs spaced emails, empty strings, and occasionally malformed JSON — patterns a fine-tuned token classifier never produces.
- **No boundary precision.** A token classifier labels each token independently giving exact span boundaries. The LLM outputs free-text strings requiring alignment back to original tokens, introducing errors.
- **Scale vs. accuracy trade-off.** LLaMA 1B (2.5B params, 2.5 GB) produced 0.42 EMAIL F1 vs. RoBERTa (124M params, 500 MB) at 0.95 F1. The 20× larger model is 53 points worse on emails without fine-tuning.

5. Recommendations

- **Address PER false negatives:** Augment training with names in quoted strings, book/song titles, and compound names ('El Hamdaoui', 'of Arc') to improve the 6.2% PER FNR.
- **Reduce PER false positives:** Add hard negative examples where capitalised words appear in non-person contexts (restaurant names, book titles, acronyms).
- **Increase max_length to 512:** The 2.5% truncation rate means some email entities at the end of long sentences are never seen. T4 GPU can handle 512 with gradient checkpointing.
- **LLM improvements:** For zero-shot, use a larger model (3B+), add JSON schema constraints, and post-process to remove empty strings and hallucinated patterns. For production PII detection, fine-tuned token classification remains strongly preferred.
- **Extend PII types:** Add phone numbers, SSNs, addresses, and credit cards using the same injection + fine-tuning pipeline.